

WASP-177b

R-2318

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-177_210908 · created 2026-08-18T00:44:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459465.739 +/- 0.0072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.087 +/- 0.047
Transit depth (Rp/Rs)^2	0.75 +/- 0.81 [%]
Orbital Inclination (inc)	89.12 +/- 2.96
Airmass coefficient 1 (a1)	1.0 +/- 0.027
Airmass coefficient 2 (a2)	-0.002 +/- 0.016
Scatter in the residuals of the lightcurve fit is	2.69 %
Best Comparison Star	#2 - [172, 427]
Optimal Method	PSF photometry
Transit Duration (day)	0.1 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.087 ± 0.047 vs 0.1618 (deviation 1.25σ)
Duration fit vs published	0.1 d vs 0.0721 d (deviation 0.95σ)
Mid-time O–C	18.7 min
Depth SNR	1.4
Residual scatter	2.69 %

Light curve

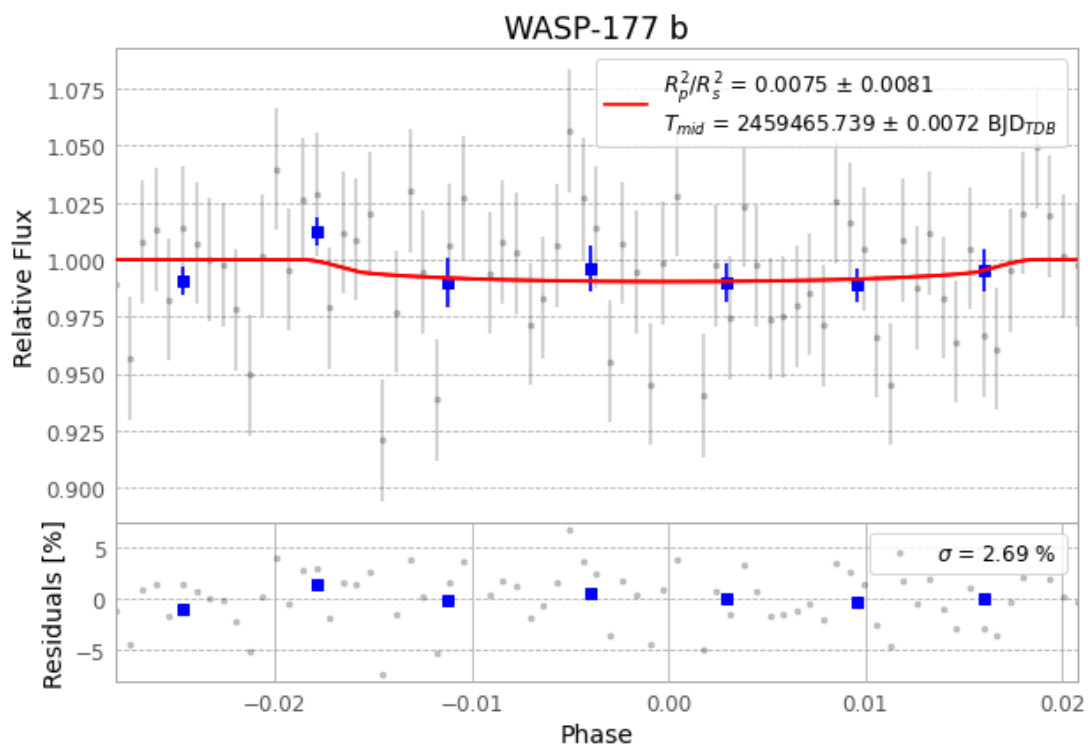


Figure 1 — FinalLightCurve_WASP-177 b_2021-09-07.png (SHA-256 3f6fc5438392b3c8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [499, 160], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 38e384e2060fa092...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

73 FITS frames (48 MB), WASP-177210908033012.FITS → WASP-177210908070612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-177-210908.log (4f0962028732...), FinalLightCurve_WASP-177 b_2021-09-07.png (3f6fc5438392...), FinalLightCurve_WASP-177 b_2021-09-07.csv (c983c4a45f77...)

Compute resources

Wall time 112.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 00:44Z.